



AW3 Application Algorithm Platform

Deployment User Manual

Industrial vision deployment for device integration, algorithm operation, and system management.

Zhejiang MRDVS Technology Co., Ltd.

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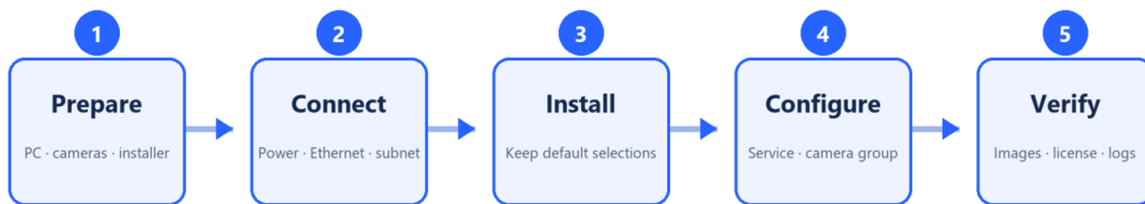
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1 Product Overview

Use AW3 to connect devices, manage camera groups, view images, manage algorithm licenses, configure protocols, and export logs. The delivered package and license determine which algorithms and features are available.

AW3 supports Volume Measurement (TorusMetric), Storage Location Detection, Pallet Docking, and Depalletizing. The delivered package and license determine which of these algorithms are available.

Deployment Workflow



2 Deployment Preparation

Preparation Item	Check
System host	Use a Windows PC or industrial computer for AW3. Make sure you have administrator access and that the computer meets the project requirements.
Camera and network	Prepare the project cameras, matching power supplies, and Ethernet cables. If you use a switch, check all connections.
Installer and license	Prepare the provided AW3 Setup package. Confirm which algorithms to install and license.
Network information	Record the IP address of each camera and of the computer adapter connected to the camera network.

3 Hardware Connection and Network Configuration

3.1 Camera Cabling

1. Connect the camera to its power supply as described in the camera hardware manual.
2. Connect the camera to the computer with an Ethernet cable. If a switch is used, connect both devices to the same switch.
3. Turn on the camera. Make sure it starts and the Ethernet cable is securely connected.

3.2 Camera IP Lookup and Subnet Requirements

To download LanxincameraViewer, open the [LanxincameraViewer download page](#) and select the Windows version under Host Software. You can also get the installer from MRDVS sales or technical support.

To find the camera IP address, install and open LanxincameraViewer on Windows. Search for cameras across subnets, select the camera, and view its IP address under Device Information.

Set the network adapter connected to the camera to the same subnet as the camera, but assign it a different IP address. You normally do not need to change the camera's IP address.

For a subnet mask of 255.255.255.0, the first three octets must match and the last octet must be different. Examples:

Computer Adapter IP	Camera IP	Result
192.168.100.250	192.168.100.90	Correct: Same subnet, different IP addresses
192.168.1.10	192.168.1.90	Correct: Same subnet, different IP addresses
10.10.10.10	10.10.10.90	Correct: Same subnet, different IP addresses
192.168.1.10	192.168.100.90	Incorrect: Different subnets
192.168.100.90	192.168.100.90	Incorrect: Duplicate IP address

All IP addresses shown above are examples. Every device on the same network must have a unique IP address.

Before changing any settings, record the original configuration and change settings only on the adapter connected to the camera. This may interrupt a remote connection. If the existing network is affected, restore the original settings.

3.3 Windows Network Adapter IP Configuration

1. Open [Settings] → [Network & internet] → [Ethernet]. Select the adapter connected to the camera, then click [Edit] next to [IP assignment].
2. Select [Manual], enable [IPv4], and enter the [IP address] and [Subnet mask]. See the examples in Section 3.2.
3. Click [Save] and confirm that the new IP address is active.

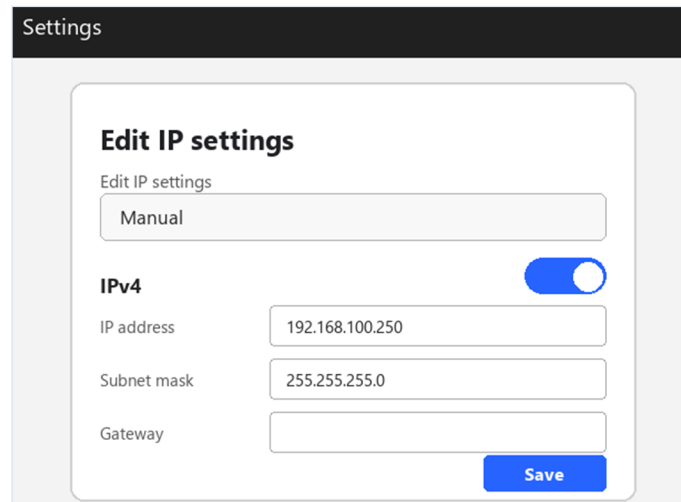


Figure 01 Windows: Enter an IPv4 Address

For a dedicated camera adapter, leave the Gateway and DNS fields blank. If the adapter also connects to another network, follow the site network requirements.

3.4 Ubuntu Network Adapter IP Configuration

1. Open [Settings] → [Network], then click the gear icon beside the wired adapter connected to the camera.

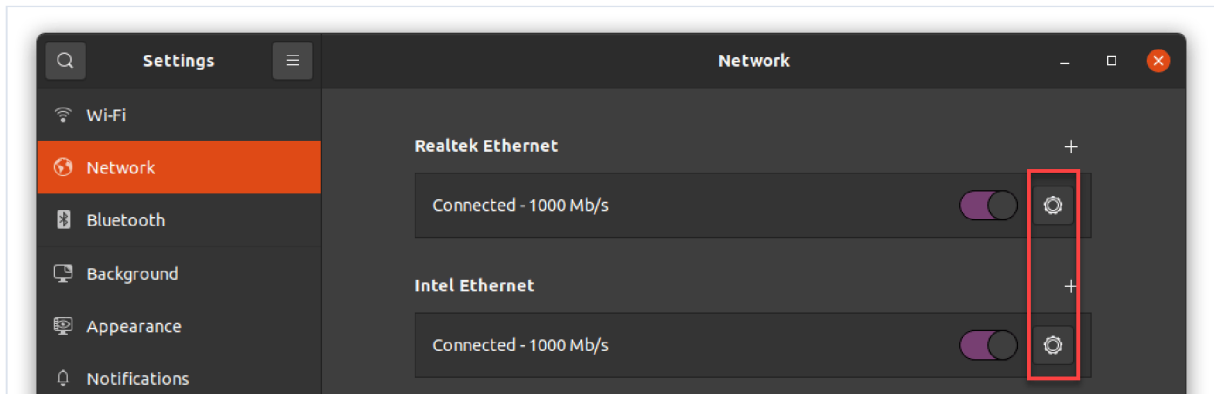


Figure 02 Ubuntu: Open the Wired Adapter Settings

2. Select [IPv4] → [Manual], then enter the [Address] and [Netmask]. See the examples in Section 3.2.
3. Click [Apply], reconnect the adapter, and confirm that the new IP address is active.

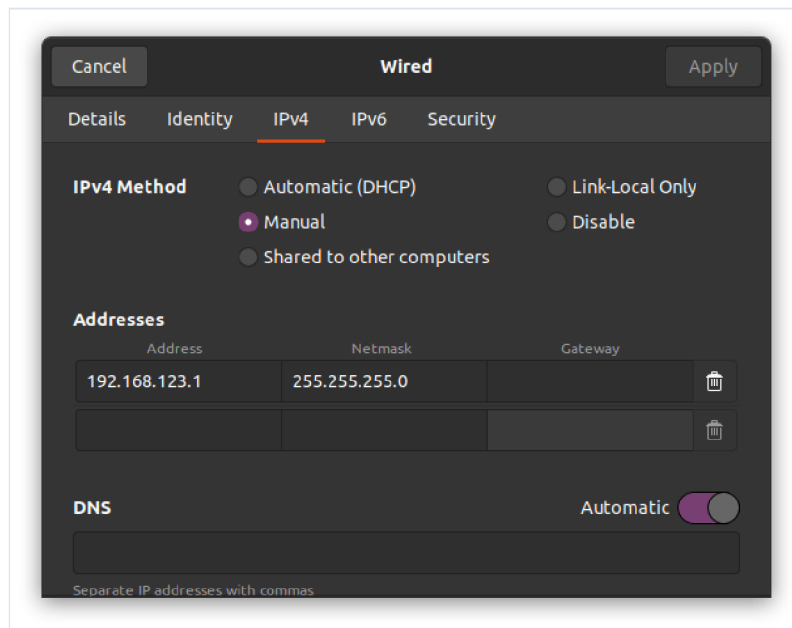


Figure 03 Ubuntu: Enter an IPv4 Address Manually

For a dedicated camera adapter, leave the Gateway and DNS fields blank. If the adapter also connects to another network, follow the site network requirements.

3.5 Camera IP Configuration

If camera IP addresses conflict or need to match the site's addressing plan, use LanxincameraViewer for Windows to change them.

1. Open LanxincameraViewer and search for the target camera.
2. Click [Others] → [Network] → [Camera IP Configuration Tool].
3. Expand the target camera entry by its device ID and record the current settings. Enter the new [IP], NetMask, and Gateway values as needed.
4. Make sure the new address is on the same subnet as the computer adapter and is not already in use. Then click [Confirm modification].

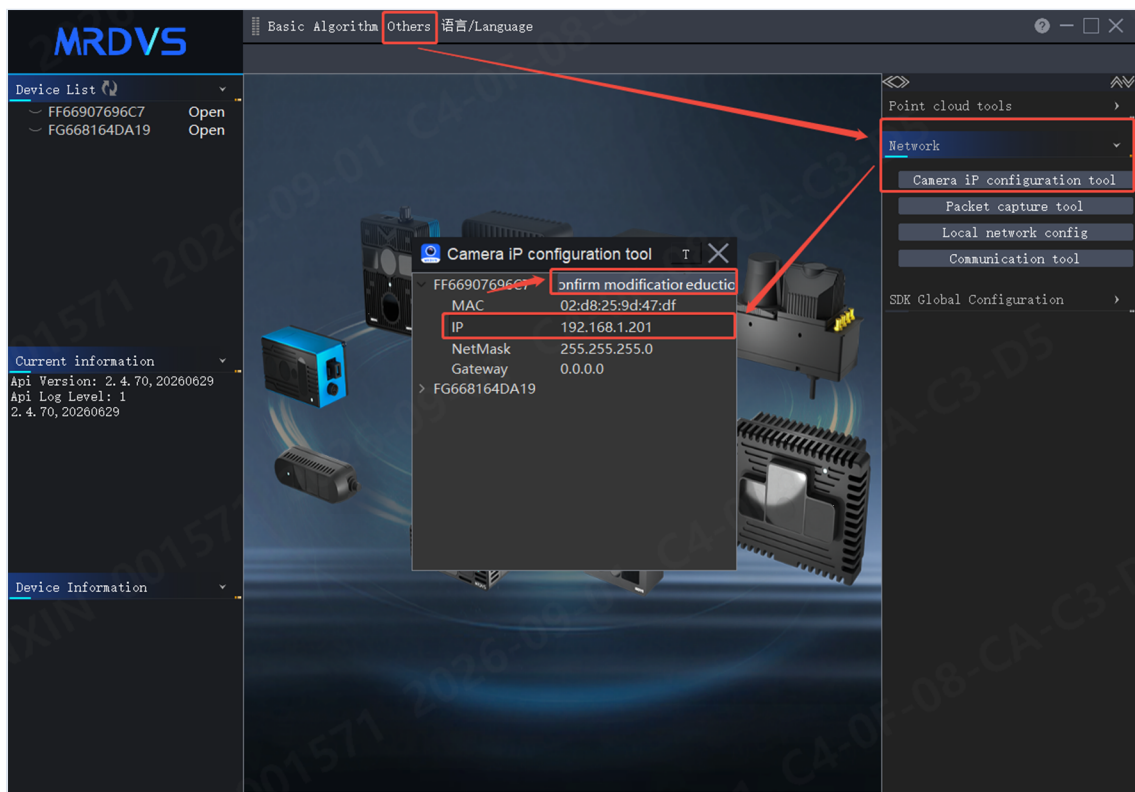


Figure 04 LanxincameraViewer: Change the Camera IP

5. Select the camera again and make sure the new IP address is active. If AW3 is already deployed, recheck the connection and image display.

When configuring multiple cameras, change one camera at a time. Record each device ID and new IP address. See Section 9.2 for network connectivity checks.

4 AW3 Installation

The installer selects all components, algorithms, and startup options by default. Keep the default selections and click [Next] on each page unless the project specifies otherwise.

4.1 Installation Components

1. Right-click the AW3 installer file and select [Run as administrator]. Select [English], then click [OK].
2. On [Select Destination Location], keep the default folder. If you choose another folder, record its location, then click [Next].
3. On [Select Components], keep all default selections and click [Next].

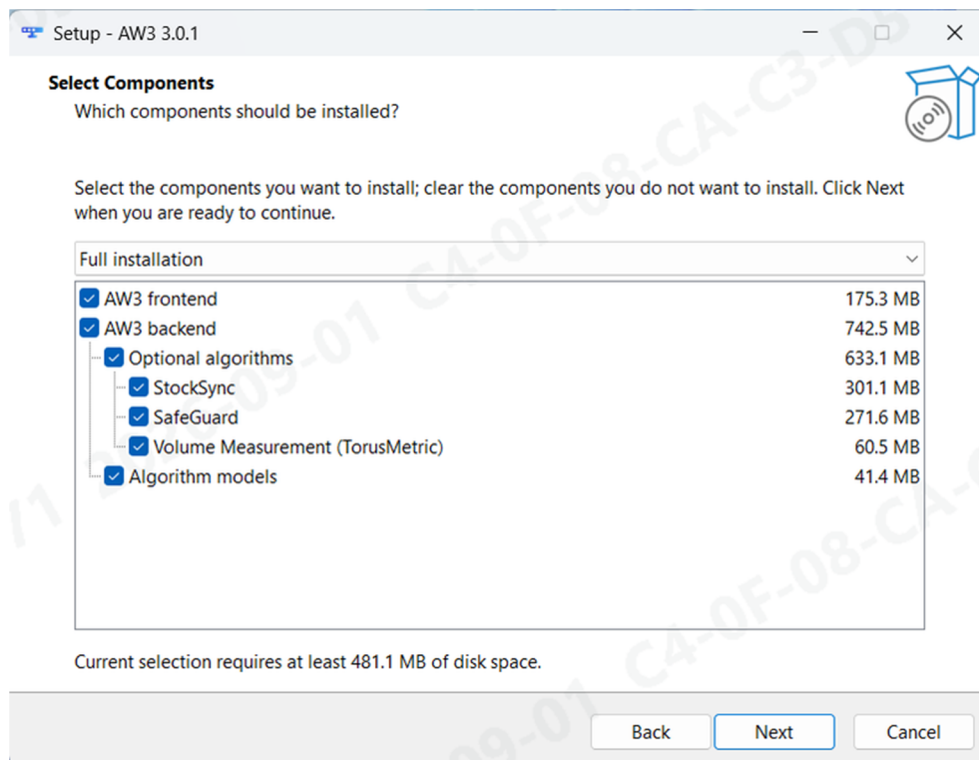


Figure 05 Default Installation Components

Component	Purpose
AW3 frontend	Interface for connecting devices, viewing data, and configuring AW3.
AW3 backend	Runs the Central Service and manages devices. Keep this component selected.
Algorithms / models	Provides the files required to run the algorithms. Keep this component selected.

The figure shows [Full installation] selected. After installation, license the algorithms as described in Chapter 6.

4.2 Installation Options and First Launch

Keep all additional tasks selected by default, then click [Next].

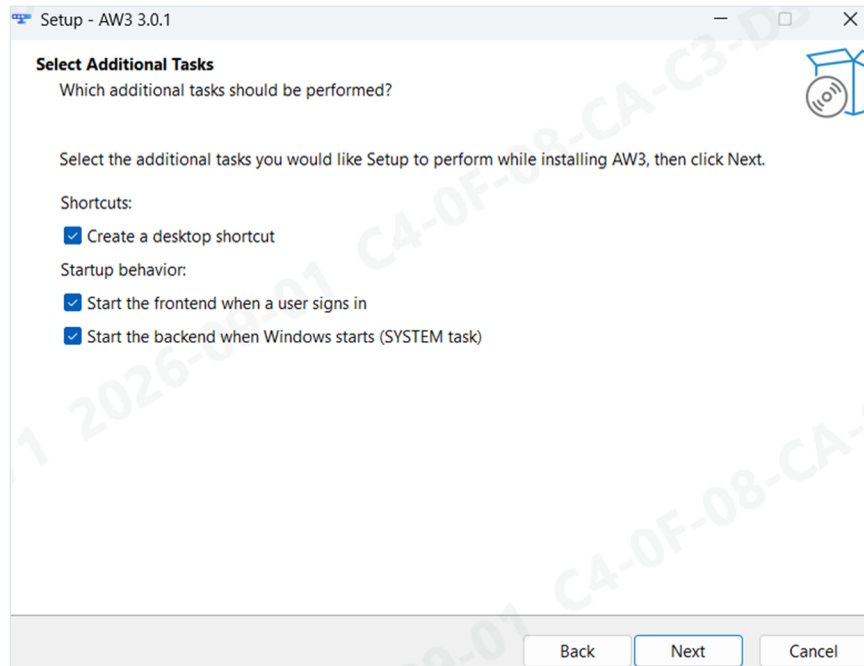
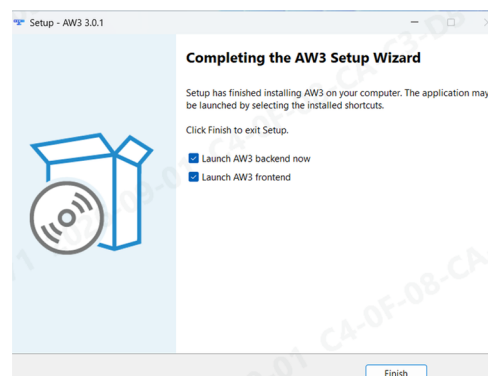


Figure 06 Desktop Shortcut and Startup Options

Option	Purpose
Create a desktop shortcut	Creates an AW3 shortcut on the desktop.
Start the frontend when a user signs in	Opens the user interface after Windows sign-in.
Start the backend when Windows starts	Starts the backend service with Windows. Keep selected.

Keep all default selections. This simplifies deployment and reduces communication and maintenance work.

Complete the Installation and Start AW3



Keep both launch options selected, then click [Finish]. To uninstall AW3, open the installation folder and run [unins000].

5 Device Connection and Image Check

5.1 Main Interface

The top bar contains the main functions. Camera groups are listed on the left, images and point clouds appear in the center, and algorithm status and results appear on the right.

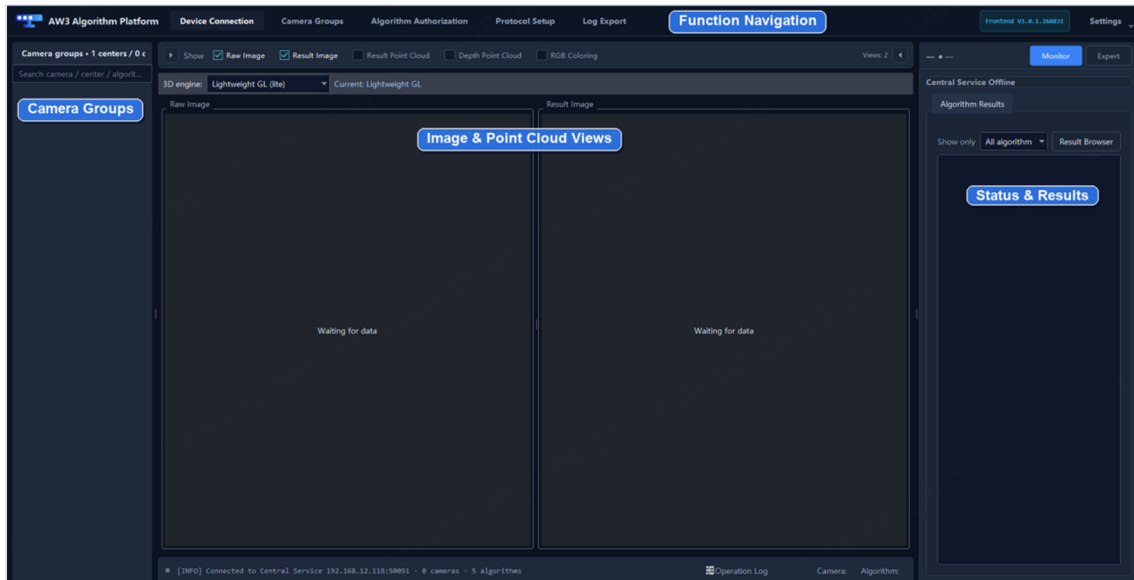


Figure 07 AW3 Main Interface

Area / Entry	Purpose
Device Connection	Check the Central Service status and find base cameras.
Camera Groups	Create and configure camera groups.
Algorithm Authorization	Generate a request key and apply a license key.
Protocol Setup	View external triggers and result forwarding settings.
Log Export	Select an algorithm, category, and time range, then export diagnostic files.
Left: Camera Group List	Select a camera group to view its data or change its settings.
Center: Image Views	View RGB images and depth point clouds.
Right: Status and Results	View algorithm status and output.

5.2 Central Service Connection and Camera Discovery

1. Click [Device Connection] on the top bar to open [Device Connection / Base Camera].
2. AW3 connects to or switches between Central Services automatically. Only when disconnected, click [Discover Devices] to check for an available Central Service.
3. Once the Central Service is connected, click [Find Base Cameras].
4. Check the camera model and [Camera IP] in the list. Click [Refresh Status] to update the status.

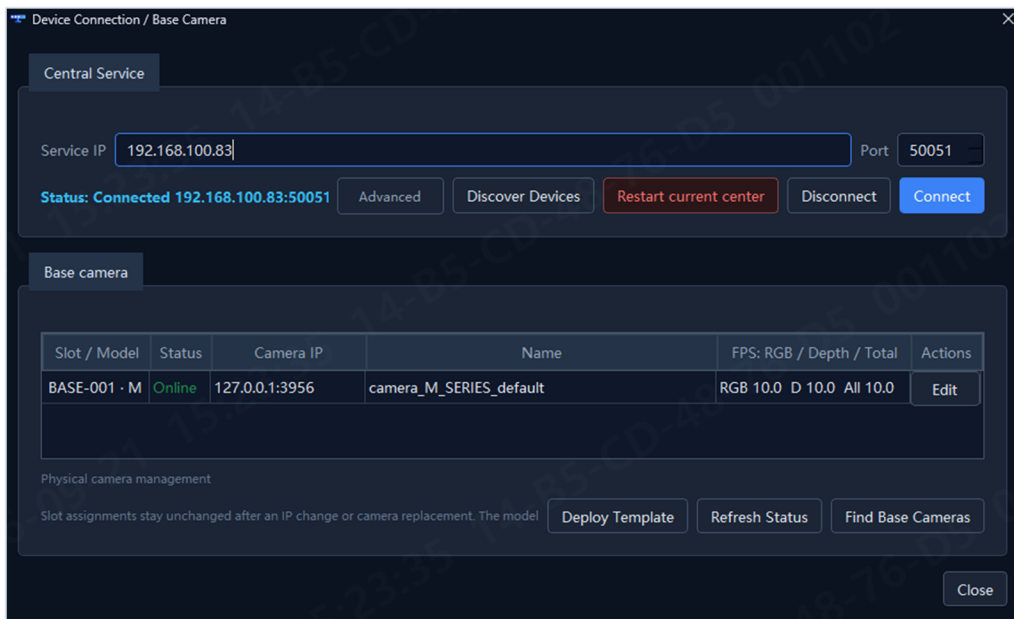


Figure 08 Central Service Connection, Restart and Camera Discovery

The addresses shown are examples. Use the values configured at the deployment site.

To verify [Service IP]: if the frontend and backend share a computer, it is the IP address of the network adapter connected to the camera. Otherwise, it is the backend computer's IP address, not the camera IP.

Completion check: The Central Service status shows Connected, and the target camera appears in the base camera list.

The Discovered status only confirms that AW3 found the device. Create and configure a camera group before checking images.

Deploy a Base Camera Template if Needed

Apply a preset camera SDK configuration to the selected base cameras. Create a camera group first so that the camera appears in the target list.

Procedure

Prerequisite: Create a camera group first so that its camera appears in the target list.

Open [Device Connection], then click [Deploy Template].

Select the template and target camera, then click [Dispatch].

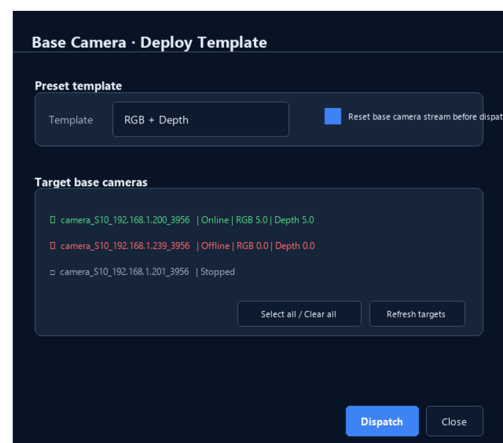


Figure 09 Base Camera Template Deployment (detail)

5.3 Create a Camera Group

A camera group is the smallest unit that can run an algorithm. It can contain one or more cameras.

1. Click [Camera Groups] on the top bar, then click [New Camera Group].
2. Enter a [Display name] and check the Central Service address. Keep the gRPC port assigned by AW3.
3. Click [Search the web], then select cameras by name or IP. You can select more than one camera.
4. Under [Run the algorithm], select the algorithm used by the project.
5. Click [Create camera group].
6. Return to the list and check the display name, physical camera IP, and algorithm. To edit an existing group, see Section 5.4.

192.168.1.250 50051 virt_1 60061

Real camera (multiple choices available; synchronized with merged calibration)

1 units selected / 1 units in total Deselect all Search the web

hoos	Name	IP	Model	Status
✓	camera_S10_192.168.1.2...	192.168.1.200:3956	—	Not connected (pending ...)

Algorithm

Run the algorithm

Volume Measurement - Unauthorized (6)

Algorithm parameter path ./vir_config/<camera_id>

Algorithm log path ./vir_config/<camera_id>/logs

Network search found 1 cameras; unmanaged items are marked "Not connected (pending registration)"

Create camera group Cancel

Figure 10 Create a Camera Group (Volume Measurement example)

Keep the parameter and log paths unchanged unless the project requires custom paths. If the algorithm status is Unauthorized, follow Chapter 6 to license it.

Completion check: The target camera group appears in the list and its name, physical camera IP, and algorithm are correct.

5.4 Modify a Camera Group

1. Click [Camera Groups] on the top bar to open the camera group list.

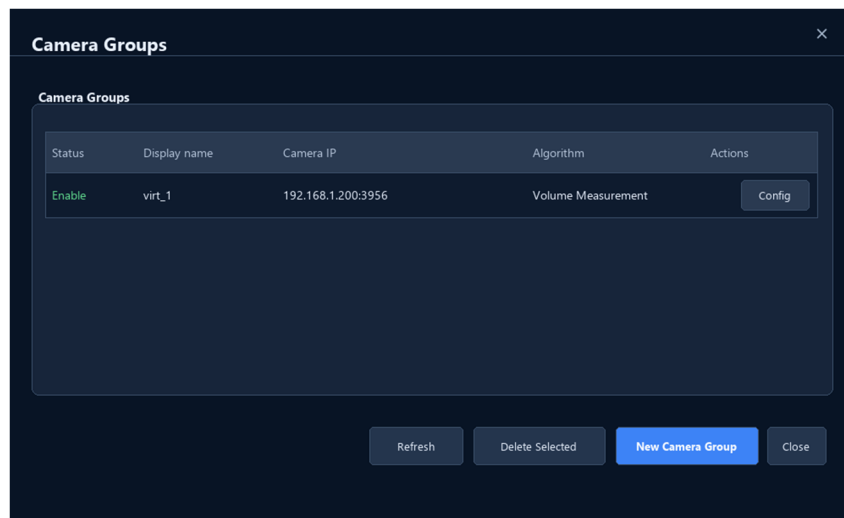


Figure 11 Camera Group List and Actions

Use [Refresh] to update the list status. [Delete Selected] removes the selected camera group. [Config] opens the group configuration.

2. Check the [Base camera]. Change the [Display name] only if needed. Keep the Config JSON unchanged unless the project specifies otherwise.

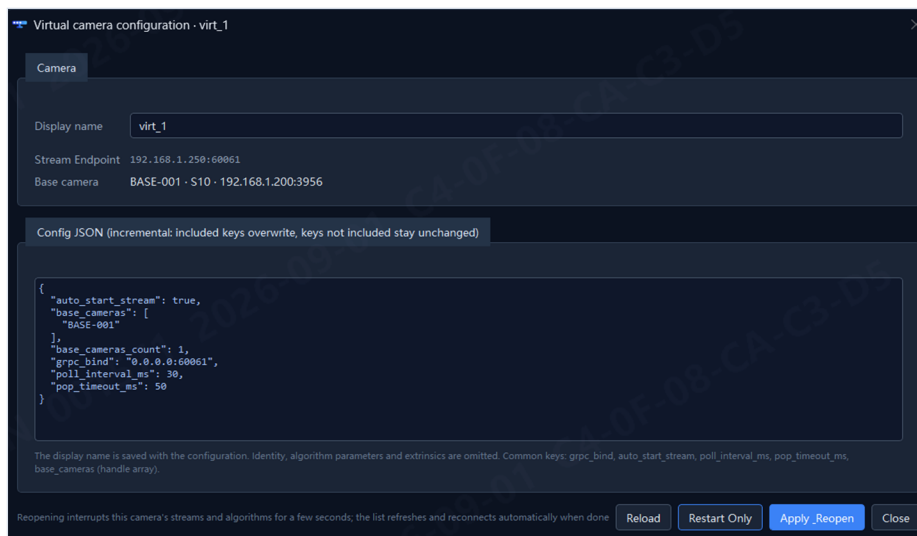


Figure 12 Existing Camera Group Configuration

3. Make sure a brief interruption is acceptable, then click [Apply_Reopen] to save and apply the change.
4. Wait for the list to refresh and reconnect. Make sure the name and camera IP are correct, then check the images as described in Section 5.5.

Caution: Reopening pauses the camera group's streams and algorithm for several seconds. Do this during an approved maintenance window.

5.5 Check the RGB Image and Depth Point Cloud

1. Close the Camera Groups window. On the left side of the main window, select the target camera group.
2. Select [Raw Image] and [Depth Point Cloud] in the display options at the top.
3. Make sure the image updates continuously and the point cloud matches the camera's field of view.

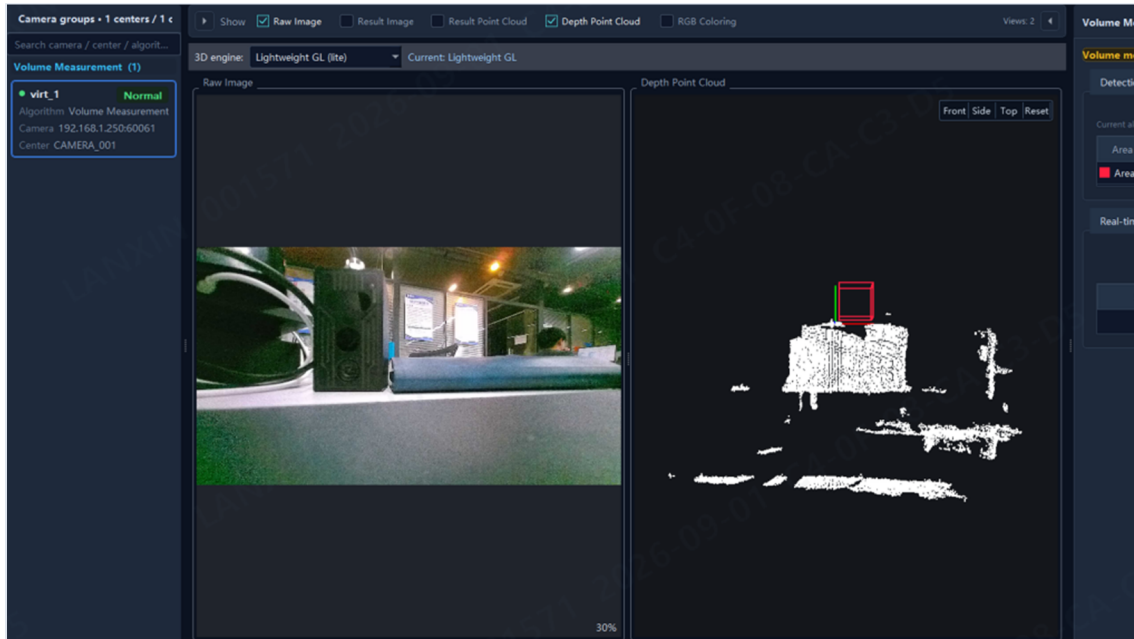


Figure 13 Camera Image and Depth Point Cloud (main interface detail)

Completion check: The device status on the left is Normal. The RGB image and depth point cloud are both visible and updating.

Caution: Seeing an image confirms only basic connectivity and data display. It does not confirm that the algorithm is licensed or that its results are valid.

If no image appears, first make sure the correct camera group is selected. Then check the Central Service connection, camera power, Ethernet cable, and subnet settings.

6 Algorithm Authorization

1. Click [Algorithm Authorization] on the top bar and select the algorithm to authorize.
2. Click [Get request key] and copy the generated authorization request key.
3. Send the request key to the project licensing contact to get the matching license key.
4. Paste the complete license key into [License key], then click [Apply license].
5. Check the [Licensed algorithms] list. If needed, click the refresh icon and check the algorithm name and validity period.

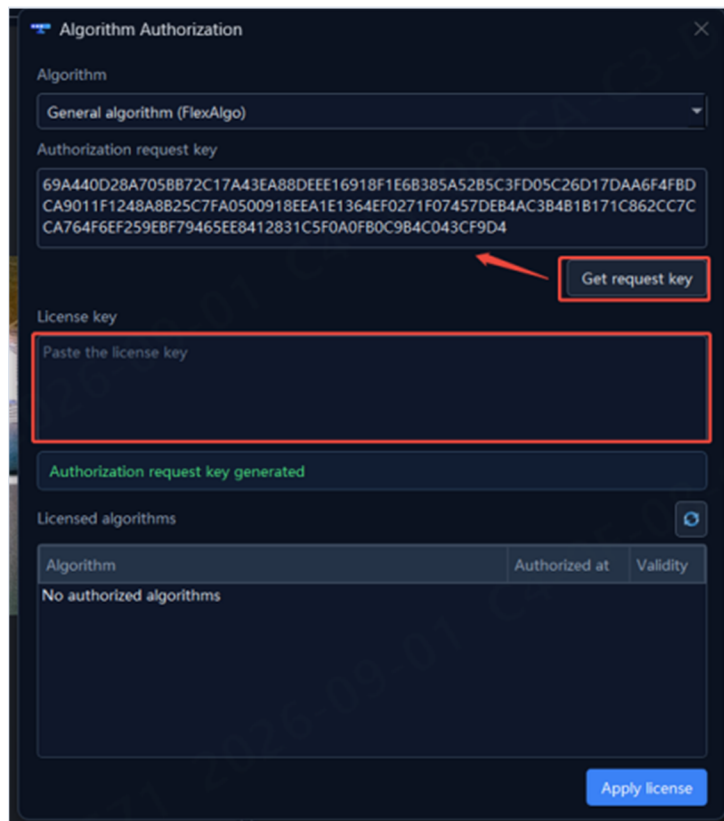


Figure 14 Algorithm Authorization Window (detail)

Completion check: The target algorithm appears in the licensed-algorithm list, and its license is valid for the required period.

Use the authorization request key to request a license, then apply the returned license key. Do not swap the two keys. After licensing, configure and validate the algorithm as described in its algorithm manual.

7 Basic System Configuration

7.1 Image Display and Rendering Settings

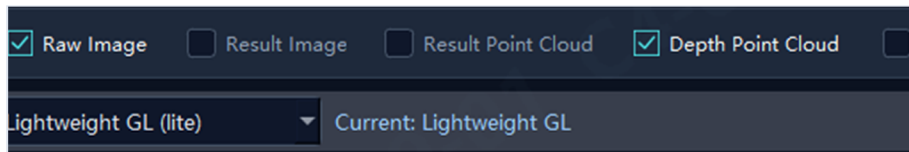


Figure 15 Display Options and 3D Engine (detail)

Option	Displayed Data
Raw Image	Color image from the camera.
Depth Point Cloud	3D depth data from the camera.
RGB Coloring	Maps RGB colors onto the depth point cloud when enabled.
Result Image / Result Point Cloud	Algorithm output. The available views depend on the algorithm and its output.
3D Engine	Renders point clouds. Keep the current setting for initial deployment.

7.2 Mode and Language Settings

Use [Monitor] and [Expert] in the upper-right corner to switch modes. Set the Expert password under [Settings].

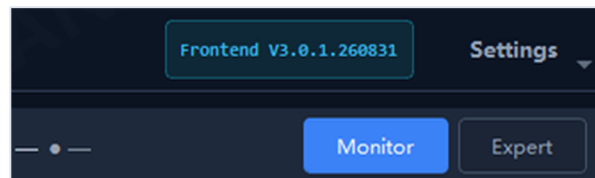


Figure 16 Monitor / Expert Mode and Settings

[Settings] lets you choose Simplified Chinese, English, Japanese, or Vietnamese.

7.3 Central Service Restart and Upgrade

Restart the Central Service

Logging in to LanxincameraViewer and opening or configuring a camera may cause Central Service streaming errors. If this occurs, open [Device Connection] and click [Restart current center] (Figure 08) to restart the AW3 backend.

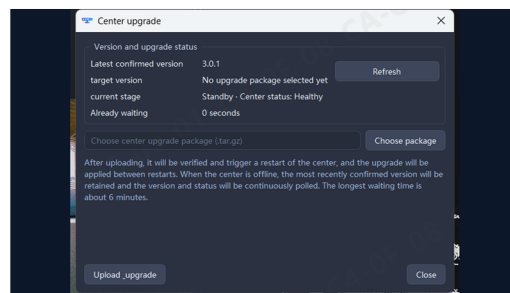
The restart briefly interrupts Central Service and camera connections. Wait for automatic reconnection, then check that images resume.

Upgrade the Central Service

Procedure

1. Click [Settings] → [Software upgrade].
2. Click [Choose package], select the .tar.gz file, verify the target version, then click [Upload_upgrade].
3. When the service returns, click [Refresh] and verify the version and Healthy status.

The package is verified before the service restarts automatically. Allow up to 6 minutes.



Center upgrade

7.4 Communication Protocol

Click [Protocol Setup] on the top bar to view external trigger requests received by AW3 and results sent to external devices.

See the manual for the selected algorithm for protocol message details. After changing a parameter, click [Save_Apply].

Function	Description
Trigger Listener	Receives a request from an external device and runs the algorithm for the specified camera group.
Result Forwarding	Sends the result to the configured external device when the algorithm finishes.

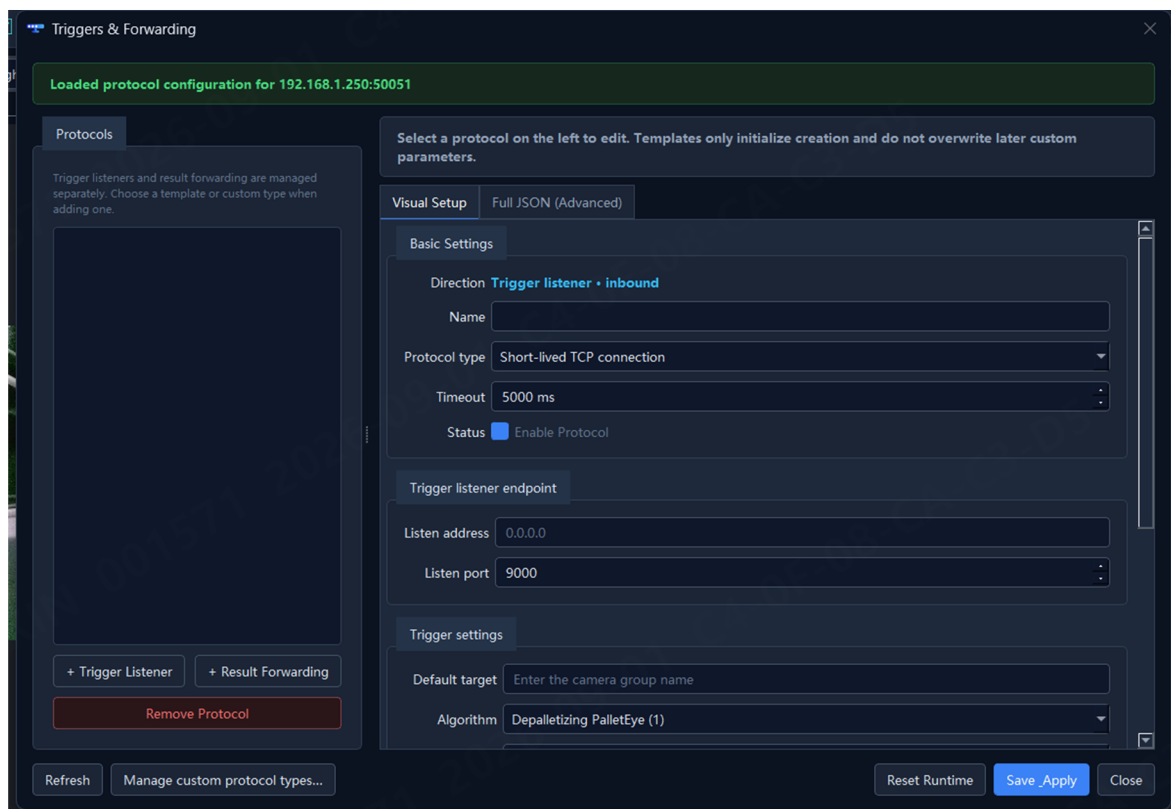


Figure 17 Triggers & Forwarding

Key Parameters

1 [Operating mode]: Clear [Use] to keep the default [push · active push] mode. AW3 sends results to the configured destination. To receive requests from an external device, select [Use] and choose [serve · service listener].

2 [Default trigger target]: This setting applies only in [serve · service listener] mode. If an incoming TRIGGER command omits target, enter the camera group display name exactly as shown in [Camera Groups]. Push mode does not use this setting.

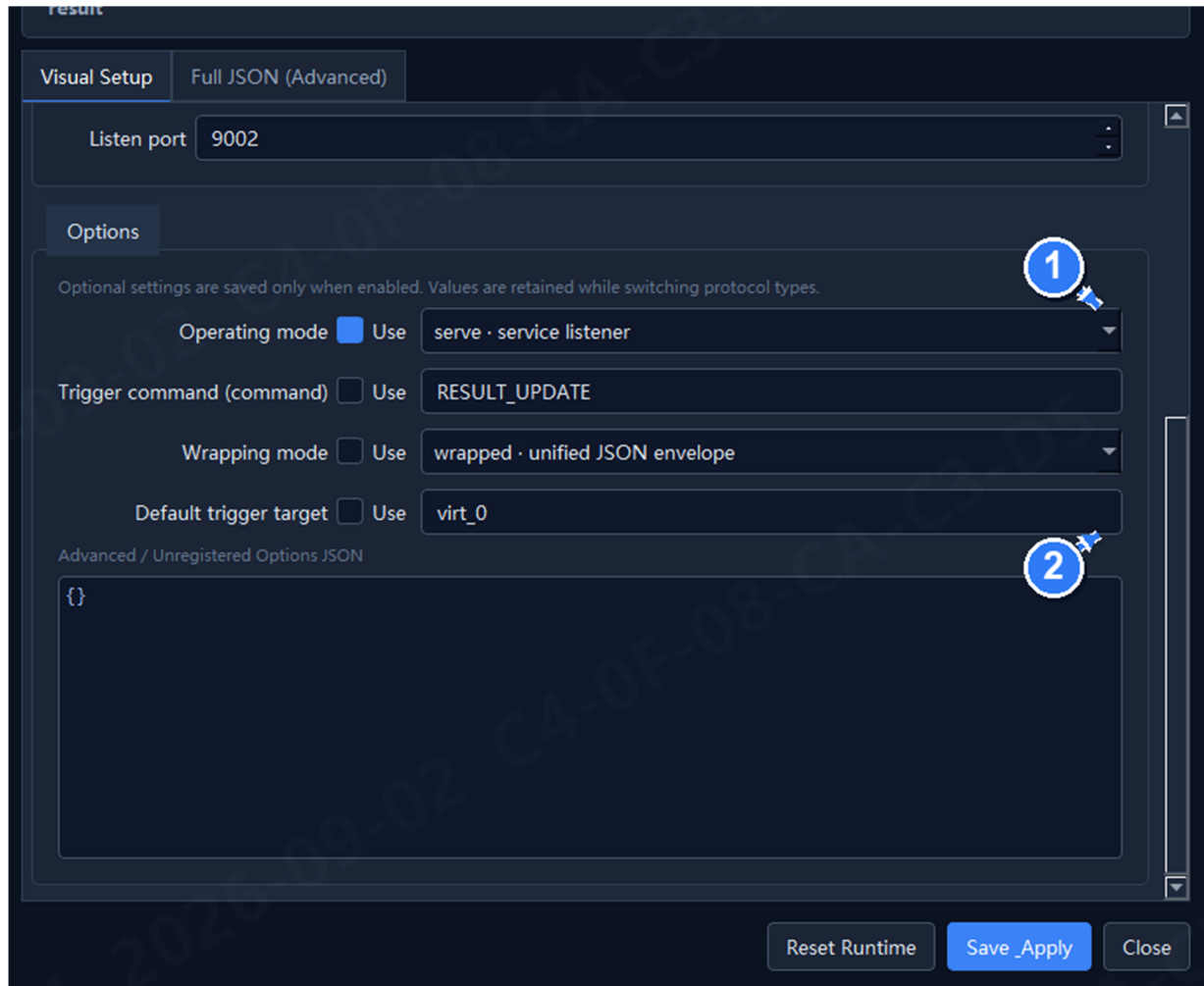


Figure 18 Operating Mode and Default Trigger Target

Usually leave [Use] clear. In [serve · service listener] mode, set [Default trigger target] only as a fallback when a TRIGGER command omits target.

8 Log and Data Export

1. Click [Log Export] on the top bar and select the algorithm associated with the issue.
2. Under [Export categories (multiple selections)], select the categories to export. You can select both Logs and Data.
3. Under [File format · Suffix (multiple selections · All selected by default)], select the formats to export. Leave [All formats] selected unless a specific format is required.
4. Select [limited] to filter by time, then enter a range that includes the time before and after the issue.
5. Click [Batch export ZIP], choose a save location when prompted, and wait for the export to finish.

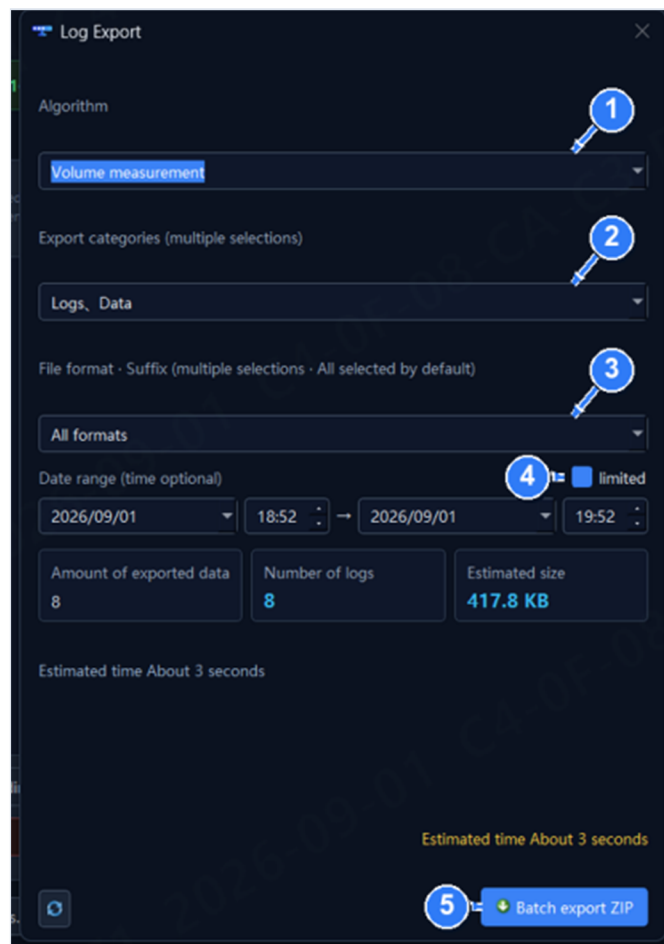


Figure 19 Log Export Window (detail)

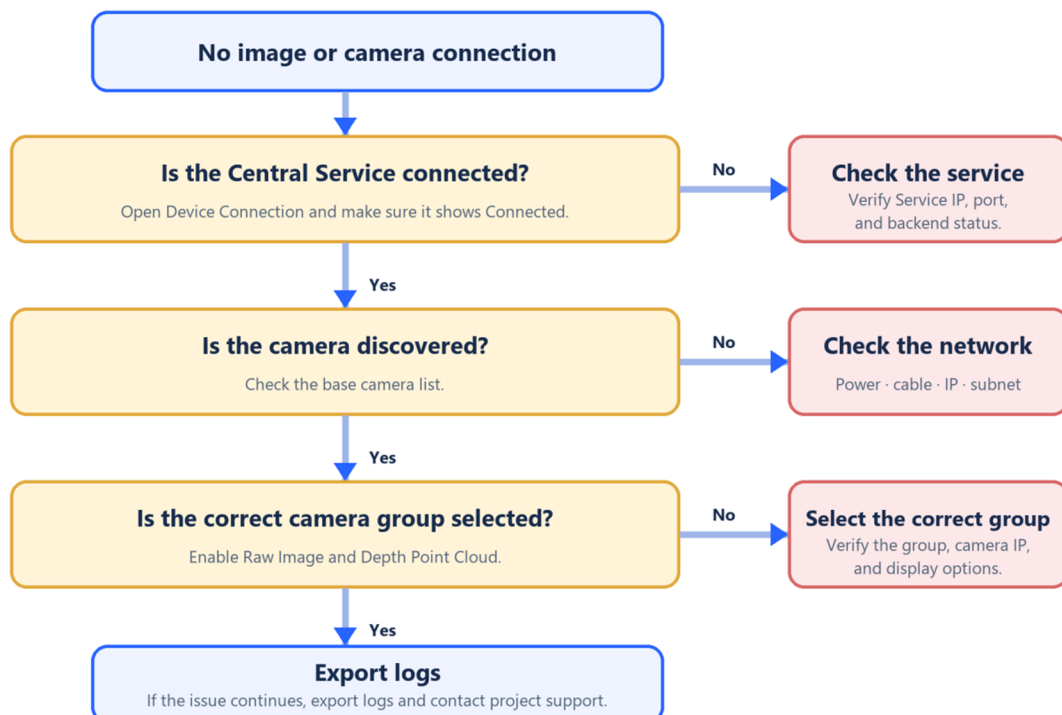
Completion check: Make sure a ZIP file was created. When reporting an issue, include the time it occurred, the software version, the target camera details, and what happened.

9 Deployment Check and Troubleshooting

9.1 Deployment Checklist

Check Item	Acceptance Criteria
Network Connection	Camera is powered on; cable is connected; computer adapter and camera are on the same subnet; IP addresses are unique.
Central Service	Device Connection status shows Connected.
Camera Group	The target camera group is listed, and its physical camera IP is correct.
Image Display	Raw Image and Depth Point Cloud are visible and updating.
Algorithm Authorization	The target algorithm has a valid license for the required period.
Final Commissioning	Follow the algorithm manual to complete calibration, parameter setup, and result validation. If external integration is required, validate the protocol as described in that manual.

Connection Troubleshooting Flow



9.2 Common Issues

Symptom	What to Check
Central Service connection failed	Check Service IP, port, backend status, and network access from the frontend to the backend host.
Camera not found	Check power and cabling. Then check the computer adapter IP, camera IP, and subnet mask.
Camera group listed but no image	Check the selected group, camera IP and enabled views. For streaming errors after using LanxincameraViewer, restart the Central Service (Section 7.3).
Image displayed but no algorithm result	Check the algorithm license and configuration. Then follow the algorithm manual to validate the results.
Authorization not applied	Make sure the algorithm and license key match and that the full key was pasted. Then refresh the licensed-algorithm list.
Export empty or failed	Check the algorithm, category, format, and time range. Make sure source data exists and the selected save location is writable.

Camera Connectivity Check (ping)

Use ping to test whether the computer can reach the camera at its IP address.

1. Windows: Press Win + R, type cmd, and press Enter. Ubuntu: Open [Terminal] from the application list.
2. Enter the command for your operating system and press Enter. Wait for all four ping attempts to finish.

Windows: ping 192.168.100.90

Ubuntu: ping -c 4 192.168.100.90

Replace the example IP address with the camera's actual IP address. Do not add a port such as :3956.

3. Check the replies and packet-loss statistics. Responses from the camera that show a response time and 0% packet loss confirm connectivity for this test.

If the request times out, the host is unreachable, or packets are lost, check camera power, the Ethernet cable, IP address, and subnet mask.

A successful ping does not confirm that images or algorithms are working. If there is no reply, the device or firewall may be blocking ICMP. Do not disable the firewall just for this test.

If the issue continues, export the logs as described in Chapter 8 and contact the project's technical support team. Avoid repeatedly changing IP addresses or unrelated settings.

Revision History and Technical Support

Version	Date	Description
V0.1	2026-08-31	Initial release covering installation, network configuration, device connection, algorithm authorization, system settings, and log export.

Technical Support

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Website: mrdvs.com

LinkedIn: [linkedin.com/company/mrdvs](https://www.linkedin.com/company/mrdvs)

Resources: [Application Algorithm Guides](#) · [LanxincameraViewer](#)